

Comparison Report of Three Machine Learning Models

This comparison focuses on the three models implemented in the research paper:

Decision Tree

Random Forest

Support Vector Machine (SVM)

The comparison is based on MAE, RMSE, R^2 , and Cross-Validation RMSE (CV_RMSE).

1. Model Performance Comparison

Model	MAE	RMSE	R^2	CV_RMSE
Decision Tree	0.368599	0.532530	-0.537508	0.541970
Random Forest	0.281927	0.395055	0.153855	0.408265
SVM	0.265217	0.398833	0.137596	0.406327

2. Decision Tree

The Decision Tree model provides a simple and interpretable approach to earthquake magnitude prediction.

Results

MAE: 0.368599

RMSE: 0.532530

R^2 : -0.537508

CV_RMSE: 0.541970

The Decision Tree has the highest MAE and RMSE and the lowest R^2 among the three models. Its negative R^2 indicates poor predictive performance on the test data.

The research paper also reports that Decision Tree performs worse than Random Forest and SVM.

Assessment: Weakest model

3. Random Forest

Random Forest is an ensemble method that combines multiple decision trees to improve prediction and reduce overfitting.

The paper uses 100 trees, which is also used in the notebook.

Results

MAE: 0.281927

RMSE: 0.395055

R^2 : 0.153855

CV_RMSE: 0.408265

Random Forest has the lowest RMSE and highest R^2 among the three models in the notebook.

The paper also identifies Random Forest as its best-performing model, with a much stronger R^2 of 0.574261.

Assessment: Best overall model

4. Support Vector Machine (SVM)

The notebook uses an RBF kernel SVR with:

kernel = 'rbf'

C = 1.0

epsilon = 0.1

gamma = 'scale'

These settings are consistent with the configuration described in the research paper.

Results

MAE: 0.265217

RMSE: 0.398833

R^2 : 0.137596

CV_RMSE: 0.406327

SVM has the lowest MAE and lowest CV_RMSE among the three notebook models. However, Random Forest has a slightly lower RMSE and higher R^2 .